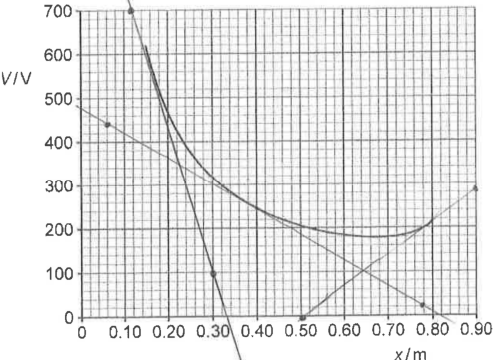
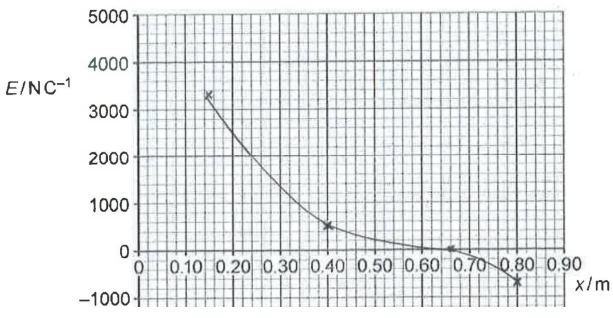


No.	Solution
4(a)	A positive charge produces a positive electric potential. A negative charge produces a negative electric potential. Since the electric potential between A and B, which is a scalar addition of electric potential due to both A and B, and it is positive throughout, both sphere are positively charged. Hence the spheres must have charges of the same sign. The sign of the charge on sphere A is positive.
4(b)	 $E = -\frac{dv}{dx} = -\text{gradient of } v - x \text{ graph}$ At $x = 0.15 \text{ m}$, $\frac{dv}{dx} = \frac{700 - 100}{0.12 - 0.30} = -3333.3 \Rightarrow E = 3330 \text{ N C}^{-1}$ At $x = 0.40 \text{ m}$, $\frac{dv}{dx} = \frac{440 - 20}{0.06 - 0.78} = -583.3 \Rightarrow E = 583 \text{ N C}^{-1}$ At $x = 0.80 \text{ m}$, $\frac{dv}{dx} = \frac{280 - 0}{0.90 - 0.50} = 700 \Rightarrow E = 700 \text{ N C}^{-1}$ 
4(c)(i)	$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ $= \frac{1}{4\pi \times 8.85 \times 10^{-12}} \times \frac{1.0 \times 10^{-8}}{0.15}$ $= 599.45 = 600 \text{ V}$

No.	Solution
4(c)(ii)	Fig.2 shows the scalar sum of electric potential due to both A and B. Since A and B are of the same signs, the electric potential in Fig.2 must be higher than (c)(i).